

Scientific Machine Learning for Adsorption of Antibiotics

Neural ODE assignment

April 24, 2025

1 Introduction

Studying the adsorption of antibiotics is crucial for several fields and has a wide range of applications. The Langmuir adsorption model, a widely used mathematical model, will be enhanced using Scientific Machine Learning techniques in this project.

2 Langmuir Adsorption Model

The Langmuir adsorption model is described by the following ordinary differential equation (ODE):

$$\frac{dq}{dt} = k_a C(Q_m - q) - k_d q \quad (1)$$

where q is the amount of antibiotic adsorbed per unit mass of the membrane, C is the antibiotic concentration in the solution, Q_m is the maximum adsorption capacity, k_a is the adsorption rate constant, and k_d is the desorption rate constant.

3 Objectives

1. Implement and run the Langmuir Adsorption ODE in Julia.
2. Generate underlying data

3. Replace the full Langmuir ODE with a Neural ODE and perform prediction.
4. Compare the Neural ODE predictions with the underlying data
5. Make a detailed and comprehensive PDF report of your observations. Make sure the report is written with sections such as Introduction, Methodology, Results and Conclusion. Make sure to add figures and tables.